# Supplementary Material

Disentangling Linguistic and Visual Compositionality Failures
in Vision-Language-Action Models

This document provides four supplementary items referenced in the main paper: **S1** Per-task success rates across all cells, **S2** Full paraphrase variant listing, **S3** Complete statistical test outputs, and **S4** Qualitative failure-mode analysis.

## Contents

# 1 Per-task Success Rates Across All Cells

Table 1 reports the success rate for every task–cell combination. Each cell uses 10 rollouts per task; C2 and C4 additionally break out per-variant SR. The CGG column shows the per-task performance drop from C1 for each stressed condition.

Table 1: Per-task success rates (SR) across all four experimental cells. $\mathrm{CGG}_\ell$, $\mathrm{CGG}_v$, $\mathrm{CGG}_f$ are per-task C1−C2, C1−C3, C1−C4 respectively. **Bold** = largest drop per column; *italic* = anomalous improvement.

| Task | C1 | C2 | C3 | C4 | $\mathrm{CGG}_\ell$ | $\mathrm{CGG}_v$ | $\mathrm{CGG}_f$ |
|---|---|---|---|---|---|---|---|
| T0: btw plate & ramekin | 0.50 | 0.37 | 0.30 | 0.60 | +0.13 | +0.20 | −0.10 |
| T1: next to ramekin | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| T2: table center | 0.40 | 0.50 | 0.20 | 0.30 | −0.10 | +0.20 | +0.10 |
| T3: on the cookie box | 0.90 | 0.50 | 0.20 | 0.20 | **+0.40** | **+0.70** | **+0.70** |
| T4: cabinet drawer | 0.10 | 0.10 | 0.00 | 0.00 | 0.00 | +0.10 | +0.10 |
| T5: on the ramekin | 0.10 | 0.07 | 0.10 | 0.20 | +0.03 | 0.00 | −0.10 |
| T6: next to cookie box | 0.70 | 0.37 | 0.50 | 0.20 | +0.33 | +0.20 | +0.50 |
| T7: on the stove | 0.50 | 0.17 | 0.50 | 0.30 | +0.33 | 0.00 | +0.20 |
| T8: next to the plate | 0.30 | 0.13 | 0.60 | 0.30 | +0.17 | *−0.30* | 0.00 |
| T9: on wooden cabinet | 0.10 | 0.07 | 0.00 | 0.10 | +0.03 | +0.10 | 0.00 |
| **Mean** | **0.360** | **0.227** | **0.240** | **0.220** | **+0.133** | **+0.120** | **+0.140** |
| **Std** | 0.283 | 0.183 | 0.218 | 0.183 | — | — | — |
| **SE** | 0.090 | 0.058 | 0.069 | 0.058 | — | — | — |

## 1.1 C2 Paraphrase Type Breakdown per Task

Table 2 shows the per-task SR for each of the three paraphrase types in Cell 2, revealing task-level variance that aggregate means hide.

Table 2: Cell 2 per-task SR stratified by paraphrase type. PCS $= 1 - \sigma$(SR across 3 variants) per task.

| Task | Synonym | Restructure | Colloquial | Mean | PCS |
|---|---|---|---|---|---|
| T0: btw plate & ramekin | 0.50 | 0.40 | 0.20 | 0.37 | 0.87 |
| T1: next to ramekin | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 |
| T2: table center | 0.70 | 0.40 | 0.40 | 0.50 | 0.84 |
| T3: on the cookie box | 0.80 | 0.60 | 0.20 | 0.53 | 0.75 |
| T4: cabinet drawer | 0.10 | 0.10 | 0.10 | 0.10 | 1.00 |
| T5: on the ramekin | 0.10 | 0.10 | 0.00 | 0.07 | 0.94 |
| T6: next to cookie box | 0.60 | 0.30 | 0.20 | 0.37 | 0.81 |
| T7: on the stove | 0.30 | 0.10 | 0.10 | 0.17 | 0.89 |
| T8: next to the plate | 0.20 | 0.10 | 0.10 | 0.13 | 0.95 |
| T9: on wooden cabinet | 0.10 | 0.10 | 0.00 | 0.07 | 0.94 |
| **Mean** | **0.34** | **0.21** | **0.13** | **0.227** | **0.90** |

## 2 Full Paraphrase Variant Listing

All 30 paraphrase variants (10 tasks × 3 types) are listed in Table 3. Variants were authored by the research team and verified to uniquely identify the correct task given full scene knowledge. Authoring guidelines: (1) synonym swap replaces only action verbs and spatial prepositions, leaving sentence structure intact; (2) restructure alters syntactic form without changing semantics (passive voice, relative clauses, participial phrases); (3) colloquial rewrite uses informal, shortened phrasing characteristic of natural human speech, often dropping articles or using everyday vocabulary.

Table 3: Complete paraphrase variants for all 10 LIBERO-Spatial tasks.

| Task | Synonym Swap | Syntactic Restructure | Colloquial Rewrite |
|---|---|---|---|
| T0 | grab the black bowl between the plate and the ramekin and put it on the plate | take the black bowl that is between the plate and the ramekin and set it on the plate | put the black bowl from between the plate and ramekin onto the plate |
| T1 | grab the black bowl beside the ramekin and put it on the plate | take the black bowl located next to the ramekin and place it onto the plate | move the bowl by the ramekin to the plate |
| T2 | grab the black bowl at the center of the table and put it on the plate | take the black bowl positioned at the table's center and set it on the plate | get the bowl from the middle of the table and put it on the plate |
| T3 | grab the black bowl on top of the cookie box and put it on the plate | take the black bowl that is resting on the cookie box and place it on the plate | move the bowl sitting on the cookie box to the plate |
| T4 | grab the black bowl inside the top drawer of the wooden cabinet and put it on the plate | take the black bowl that is in the top drawer of the wooden cabinet and set it on the plate | get the bowl out of the top cabinet drawer and put it on the plate |
| T5 | grab the black bowl on top of the ramekin and put it on the plate | take the black bowl that is sitting on the ramekin and place it on the plate | move the bowl on the ramekin over to the plate |
| T6 | grab the black bowl beside the cookie box and put it on the plate | take the black bowl that is positioned next to the cookie box and set it on the plate | get the bowl near the cookie box and put it on the plate |
| T7 | grab the black bowl on the stove and put it on the plate | take the black bowl that is sitting on the stove and place it on the plate | move the bowl off the stove and onto the plate |

*Continued on next page*

| Task | Synonym Swap | Syntactic Restructure | Colloquial Rewrite |
|---|---|---|---|
| T8 | grab the black bowl beside the plate and put it on the plate | take the black bowl that is next to the plate and set it on top of the plate | move the bowl next to the plate onto the plate |
| T9 | grab the black bowl on the wooden cabinet and put it on the plate | take the black bowl that is resting on the wooden cabinet and set it on the plate | move the bowl from the wooden cabinet to the plate |

## 3 Complete Statistical Test Outputs

### 3.1 Wilcoxon Signed-Rank Test: C1 vs C2 (Linguistic Stress)

**Hypothesis:** $H_0$: no difference in per-task SR between C1 and C2.

Table 4: Per-task signed differences (C1−C2) used in the Wilcoxon test.

| Task | C1 SR | C2 SR | Diff | Note |
|---|---|---|---|---|
| T0 | 0.50 | 0.37 | +0.13 | |
| T1 | 0.00 | 0.00 | 0.00 | tie (excluded) |
| T2 | 0.40 | 0.50 | −0.10 | reversal |
| T3 | 0.90 | 0.50 | +0.40 | largest drop |
| T4 | 0.10 | 0.10 | 0.00 | tie (excluded) |
| T5 | 0.10 | 0.07 | +0.03 | |
| T6 | 0.70 | 0.37 | +0.33 | |
| T7 | 0.50 | 0.17 | +0.33 | |
| T8 | 0.30 | 0.13 | +0.17 | |
| T9 | 0.10 | 0.07 | +0.03 | |
| Non-zero differences ($n_{\text{eff}}$) | | | $n_{\text{eff}} = 8$ | |
| Wilcoxon statistic $W$ | | | $W = 34$ | |
| Two-tailed $p$-value | | | $p = 0.036$ * | |

**Interpretation:** The linguistic perturbation produces a statistically significant reduction in SR at $\alpha = 0.05$. One reversal (T2: −0.10) and two ties (T1, T4) are present; effective $n = 8$.

### 3.2 Wilcoxon Signed-Rank Test: C1 vs C3 (Visual Stress)

Table 5: Per-task signed differences (C1−C3) used in the Wilcoxon test.

| Task | C1 SR | C3 SR | Diff | Note |
|---|---|---|---|---|
| T0 | 0.50 | 0.30 | +0.20 | |
| T1 | 0.00 | 0.00 | 0.00 | tie (excluded) |
| T2 | 0.40 | 0.20 | +0.20 | |
| T3 | 0.90 | 0.20 | +0.70 | largest drop |
| T4 | 0.10 | 0.00 | +0.10 | |
| T5 | 0.10 | 0.10 | 0.00 | tie (excluded) |
| T6 | 0.70 | 0.50 | +0.20 | |
| T7 | 0.50 | 0.50 | 0.00 | tie (excluded) |
| T8 | 0.30 | 0.60 | −0.30 | anomalous improvement |
| T9 | 0.10 | 0.00 | +0.10 | |
| Non-zero differences ($n_{\text{eff}}$) | | | $n_{\text{eff}} = 7$ | |
| Wilcoxon statistic $W$ | | | $W = 26$ | |
| Two-tailed $p$-value | | | $p = 0.177$ (ns) | |

**Interpretation:** The visual effect does not reach significance. The anomalous T8 improvement (−0.30) and three ties reduce $n_{\text{eff}}$ to 7, substantially limiting power. A power analysis indicates $n \approx 20$ tasks would be needed to detect a true effect of $\delta = 0.12$ at 80% power with $\alpha = 0.05$.

## 3.3 Wilcoxon Signed-Rank Test: C1 vs C4 (Full Stress)

Table 6: Per-task signed differences (C1−C4) used in the Wilcoxon test.

| **Task** | **C1 SR** | **C4 SR** | **Diff** | **Note** |
|---|---|---|---|---|
| T0 | 0.50 | 0.60 | −0.10 | reversal |
| T1 | 0.00 | 0.00 | 0.00 | tie (excluded) |
| T2 | 0.40 | 0.30 | +0.10 | |
| T3 | 0.90 | 0.20 | +0.70 | largest drop |
| T4 | 0.10 | 0.00 | +0.10 | |
| T5 | 0.10 | 0.20 | −0.10 | reversal |
| T6 | 0.70 | 0.20 | +0.50 | |
| T7 | 0.50 | 0.30 | +0.20 | |
| T8 | 0.30 | 0.30 | 0.00 | tie (excluded) |
| T9 | 0.10 | 0.10 | 0.00 | tie (excluded) |
| Non-zero differences ($n_{\text{eff}}$) | | | $n_{\text{eff}} = 7$ | |
| Wilcoxon statistic $W$ | | | $W = 23$ | |
| Two-tailed $p$-value | | | $p = 0.128$ (ns) | |

## 3.4 CGG Decomposition Summary

Table 7: CGG decomposition with standard errors and significance.

| **Metric** | **Value** | **SE** | ***p*-value** | **Sig.** |
|---|---|---|---|---|
| $\text{CGG}_{\text{ling}}$ | +0.133 | 0.060 | 0.036 | * |
| $\text{CGG}_{\text{vis}}$ | +0.120 | 0.069 | 0.177 | ns |
| $\text{CGG}_{\text{full}}$ | +0.140 | 0.058 | 0.128 | ns |
| $\text{CGG}_{\text{syn}}$ | −0.113 | — | — | sub-additive |

The negative $\text{CGG}_{\text{syn}}$ indicates **sub-additive** failure: combining both stressors is less damaging than the sum of their individual effects (expected combined drop $= 0.133 + 0.120 = 0.253$; observed $= 0.140$). This is consistent with a *shared bottleneck* interpretation: the tasks that are hardest under linguistic stress (T1, T4, T9 already at floor) cannot degrade further under combined stress, compressing the combined effect.

# 4 Qualitative Failure Mode Analysis

## 4.1 Task Cluster Characterisation

We identify three behavioural clusters based on patterns in Table 1:

### Cluster A — Floor Tasks (T1, T4, T9)

These tasks achieve SR $\leq 0.10$ in C1 and remain at or near 0.00 across all cells. Their failures are unaffected by either linguistic or visual perturbation, indicating the failure mode is in *visual grounding* of the reference object, not instruction parsing.

- **T1 (next to ramekin):** The ramekin is a small, visually similar object to the bowl target. The model appears to confuse target and reference, grasping the ramekin itself or missing both. No perturbation changes this outcome.
- **T4 (cabinet drawer):** Requires the model to locate a closed drawer, a spatial relationship involving a 3D container not well-represented in the training distribution. All rollouts exhaust 300 steps without grasping.
- **T9 (wooden cabinet):** The "wooden cabinet" reference is visually ambiguous (similar texture to parts of the table). The model consistently reaches toward incorrect locations.

### Cluster B — Robust Tasks (T6, T7)

T6 (next to cookie box) and T7 (stove) maintain SR $\geq 0.50$ under visual stress and $\geq 0.17$ under full stress. The cookie box and stove are visually distinctive and spatially salient reference objects, enabling reliable visual grounding even under pose perturbation. T7 shows zero linguistic degradation under synonym swap, suggesting "stove" has very consistent representation in the model's language encoder.

### Cluster C — Fragile Tasks (T3, T6 under C4, T8)

- **T3 (on the cookie box):** Highest baseline SR (0.90) but largest absolute CGG under visual stress ($-0.70$, C1$\rightarrow$C3). Hypothesis: the model has a highly tuned policy for this specific cookie-box configuration that is disrupted by even small pose changes in the camera perspective, causing it to miss the grasp.
- **T8 (next to the plate) anomalous improvement:** SR rises from 0.30 (C1) to 0.60 (C3) under visual perturbation. Hypothesis: the canonical robot pose creates a visual ambiguity between the bowl-next-to-plate and the plate itself (two similarly sized flat objects side by side); pose perturbation shifts the viewing angle so the bowl becomes more visually distinct. This is the only case where visual perturbation is *beneficial*, and it highlights that "visual stress" is not uniformly detrimental—it depends on whether the original pose was already suboptimal for the model.
- **T2 (table center) linguistic reversal:** SR rises from 0.40 (C1) to 0.50 (C2, synonym swap) and 0.70 under synonym swap specifically. Hypothesis: the original instruction "pick up the black bowl from the table center" may be suboptimally phased relative to training data; the synonym "grab the black bowl at the center of the table" better matches a seen training phrasing.

## 4.2 Failure Taxonomy

Based on the above analysis, we propose the following failure taxonomy for OpenVLA on LIBERO-Spatial:

Table 8: Failure taxonomy for OpenVLA on LIBERO-Spatial.

| **Failure type** | **Root cause** | **Affected tasks** | **Signature** |
|---|---|---|---|
| Visual grounding failure | Reference object visually ambiguous or OOD | T1, T4, T9 | SR = 0 regardless of language condition |
| Linguistic formality sensitivity | Instruction too informal relative to training distribution | T3, T6, T7 (colloquial) | Large SR drop under colloquial only |
| Pose-dependent policy | Learned policy over-fits canonical camera angle | T3 (C3), T8 (C3 improves) | SR changes sharply under small pose perturbation |
| Instruction phrasing mismatch | Canonical instruction suboptimal vs. training data | T2 (C2 synonym improves) | SR improves under synonym swap |

### 4.3 Rollout Behaviour Observations

In all failure cases across all cells, failed rollouts run to the full 300-step budget with the arm continuing to reach toward incorrect locations or oscillating near the bowl without grasping. This is consistent with the absence of a confidence or progress signal in the standard OpenVLA architecture: the model has no mechanism to detect that its current trajectory is non-viable and does not abort or retry. Future work should explore early-termination heuristics or uncertainty-based replanning to reduce wasted compute on inevitable failures.